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Deep Learning based medical image segmentation — Part 1 — Getting the data



Diya S

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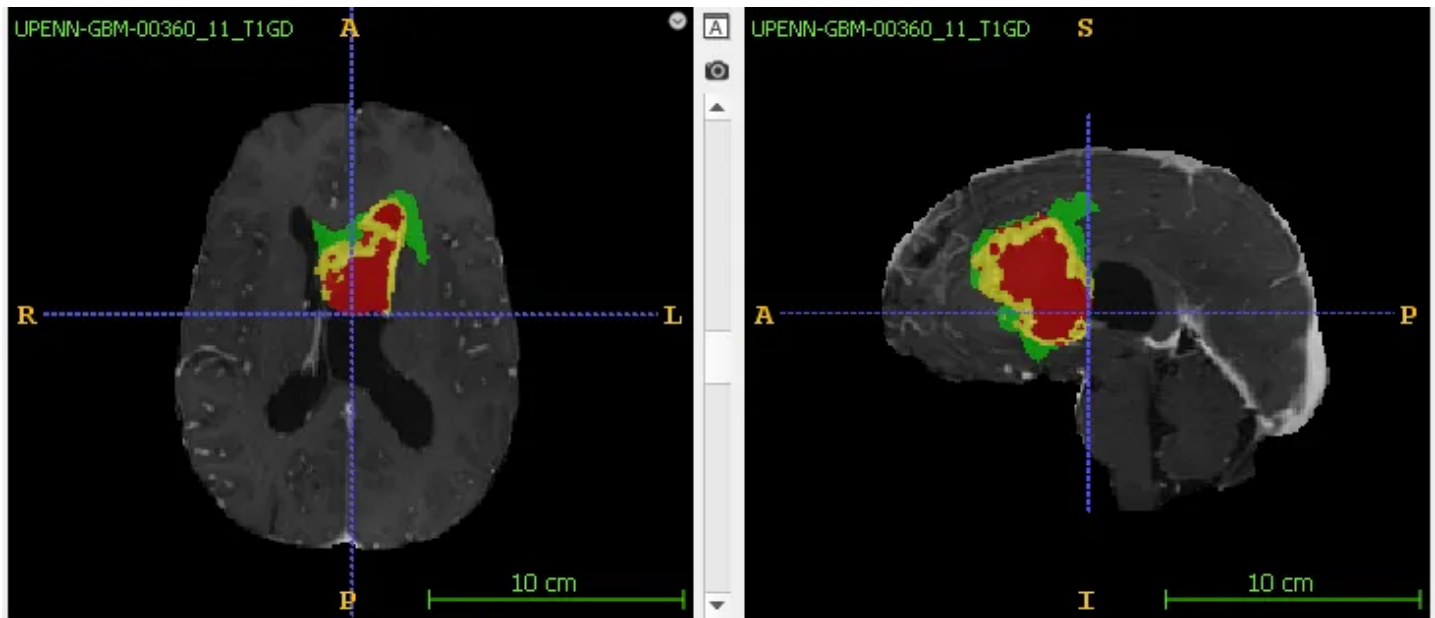
5 min read · May 14, 2023



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Introduction



Tumor Segmentation in ITK-SNAP. Image by author

The goal of brain tumor segmentation is to accurately detect and mark the location and extent of the tumor tissue from normal tissue so accurate surgical treatment can be performed. Since Glioblastoma is fast growing, there may not be a clear demarcation between healthy and tumorous tissue. In addition, brain tumors have sub-regions such as enhancing tumor and edema (swelling) that will also need to be clearly segmented¹. 3D-MRI scans with multiple modalities (or parameters) are used to provide a spatial view that helps in accurate tumor segmentation. In this study, the authors have provided the following MRI modalities² — T1 (spin lattice relaxation), T1-GD (also called T1 post contrast, T1 contrast enhanced), T2 (spin-spin relaxation), and T2-FLAIR (Fluid Attenuation Inversion Recovery).

How can we use AI — in this case Deep Learning — to segment brain tumor MRI scans? In this post, we will look at downloading and analyzing a Glioblastoma brain tumor dataset from TCIA. If you want a description of

TCIA and other cancer related data portals, read my earlier [post here](#). In future posts, I will walk through how to train and test a segmentation model from scratch. I use Google Drive and Colab for all my training, so the directions below pertain to these platforms only.

Step 1 — Download the data

- [Link to TCIA UPenn-GBM data download page](#)
- [Link to research paper describing the study](#)

I recommend downloading the NIfTI³ files as they are relatively easier to work with for a beginner. It is also smaller in size compared to the DICOM⁴ files. Finally, we want to work with 3D-MRI data, so let's *not* use the histopathology images for this exercise.

Data Access

Data Type	Download all or Query/Filter	License
Images (DICOM, 139.4 GB)	  (Download requires the NBIA Data Retriever)	CC BY 4.0
Images (NifTI, 69 GB)	 (Download and apply the IBM-Aspera-Connect plugin to your browser to retrieve this faspex package)	CC BY 4.0
Histopathology Images (NDPI, 149 GB)	  (Download and apply the IBM-Aspera-Connect plugin to your browser to retrieve this faspex package)	CC BY 4.0
Clinical Data (CSV, 51 kB)		CC BY 4.0
Histopathology to Radiology Filename Mapping (CSV, 2 kB)		CC BY 4.0

Let's first download the multiparametric MRI data. You'll need to install the IBM Aspera Plugin in your browser to download the dataset. This plugin will allow you to pause and resume downloads if needed. When you click on the Download button, Aspera plugin will launch and ask to be added as a browser extension if you are using Chrome.

The screenshot shows the Cancer Imaging Archive (CICA) interface. A modal dialog box titled "Required Steps" is open, instructing the user to install or upgrade to IBM Aspera Connect 3.11.2 to enable uploads and downloads. The dialog lists two steps: Step 1 (marked with a green checkmark and a puzzle piece icon) is "Install Extension", and Step 2 (marked with a download icon) is "Install Connect". Below the steps, there is a link "Already installed? Refresh / Troubleshoot". In the background, the CICA interface shows a package titled "UPENN-GBM NIfTI files" with a "Browse and Download Contents" button. Below this, a table lists the package contents, including a folder named "NIfTI-files".

Required Steps

To enable Aspera uploads and downloads, you need to install or upgrade to IBM Aspera Connect 3.11.2.

Step 1  [Install Extension](#)

Step 2  [Install Connect](#)

Already installed? [Refresh](#) / [Troubleshoot](#)

Package - UPENN-GBM NIfTI files

From: Natasha Honomichl
To: help@cancerimagingarchive.org
CC (on download): Natasha Honomichl
Date Sent: 23 Jun 2022 10:06 AM

Note: Supporting data NIfTI files (GBM) patients from the

Browse and Download Contents

Path: / PKG - UPENN-GBM NIfTI files

Name	Validation Error?	Size
NIfTI-files		

Select: [All](#) , [None](#)

Once the Aspera extension is installed, click on the NiFTI-files folder to browse the package contents. We don't need to download the entire package for this project, so let's save some download time and space. Let's see what we have in the package -

- **automated_seg** — contains tumor segmentation files generated automatically by the study's authors, presumably through the software packages `dfs`. Each patient has one "automated approximate segmentation" file. Later, we can compare our segmentation file with this dataset to see how our model fares.
- **images_DSC** — contains MRI scans collected using the Dynamic Susceptibility Contrast (DSC) technique. Each patient has MRI scans for DSC, relative Peak Height (PH), Percent Signal Recovery (PSR), and relative Cerebral Blood Volume (rCBV). We will not use this dataset for

our project.

- **images_DTI** — contains MRI scans collected using the Diffusion Tensor Imaging (DTI) technique. Each patient has MRI scans that assess white matter (WM) sensitivity through different metrics such as Repetition Time (TR), Radial Diffusivity (RD), Fractional Anisotropy (FA) and Axial Diffusivity (AD). We will not use this dataset for our project.
- **images_seg** — contains tumor segmentation files, one for each patient, manually segmented by expert radiologists. This will be our ground truth (GT) dataset that we will compare our AI generated segmentation files to.
- **images_structural** — contains the skull stripped images. **This will be our primary dataset.** It has one sub-folder for each of the 630 patients that were part of the UPenn GBM study. Under each patient's sub-folder, there are 4 NIfTI files (T1, T1GD, T2 and FLAIR), each corresponding to one MRI mode.
- **images_structural_unstripped** — contains raw, skull unstripped MRI data, one sub-folder for each patient. We want to run our segmentation model training on skull stripped images, so we can ignore this folder.

Browse and Download Contents or [Download Entire Package](#)

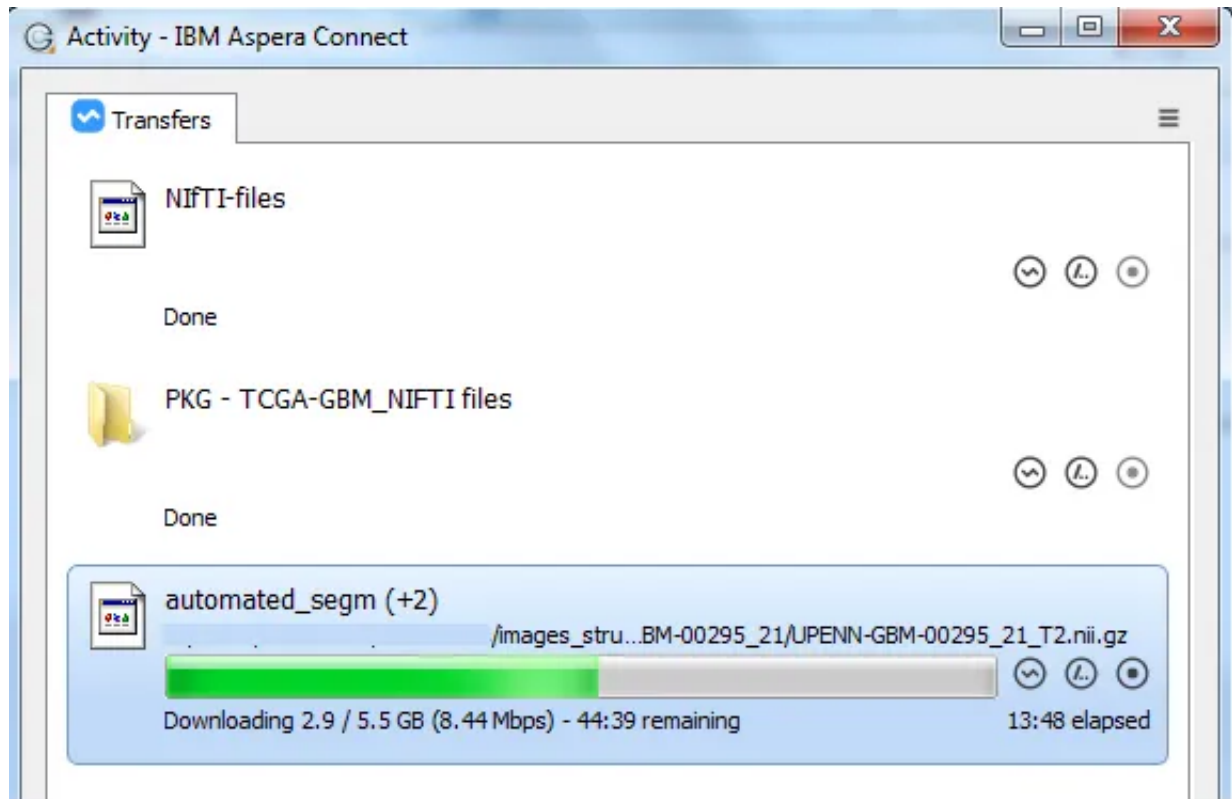
Path: / [PKG - UPENN-GBM NIfTI files](#) / NIfTI-files

	Name
☒	 automated_segm
☐	 images_DSC
☐	 images_DTI
☒	 images_segm
☒	 images_structural
☐	 images_structural_unstripped
☐	  .DS_Store

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- Go ahead and select these 3 folders and click the download button — **automated_segm**, **images_segm**, **images_structural**. The size is around 5.5 GB. Based on your internet speed, this will take a while. It took me several retries to finish downloading the files.



Don't forget to go back to the main TCIA UPenn GBM collection page and download the clinical data file as well.

- **UPENN-GBM_clinical_info_v1.0.csv** — Comma Separated Values (CSV) file with patient clinical information such as age, gender, genomics, extent of resection. We may potentially use this data later to see if we can conduct survival analysis.

Step 2 — Upload your data to Google drive

You can skip this step if you're running the training on your local machine. The UPenn GBM NIfTI dataset is 69 GB in size. I first downloaded the data to my laptop, unzipped it, and then uploaded it to my Google Drive. In my example below, I created a folder called UPENN-GBM and uploaded the

files there. You can create a folder with any name you like, but you will need to reference your particular file path in your code.

My Drive > TCIA > UPENN-GBM ▾



File type ▾

People ▾

Last modified ▾

Name ↑	Owner	Last modified ▾	File size
automated_seg	me	10:53 AM	—
images_seg	me	10:53 AM	—
images_structural	me	10:53 AM	—

3 uploads complete ▾ ×

images_structural	3355 of 3355	✓
automated_seg	611 of 611	✓
images_seg	147 of 147	✓

Next week, we will analyze the dataset and determine our approach to automated segmentation using deep learning. Stay tuned.

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Written by Diya S

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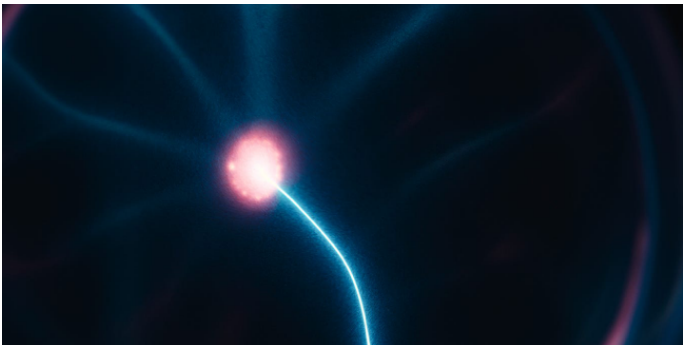
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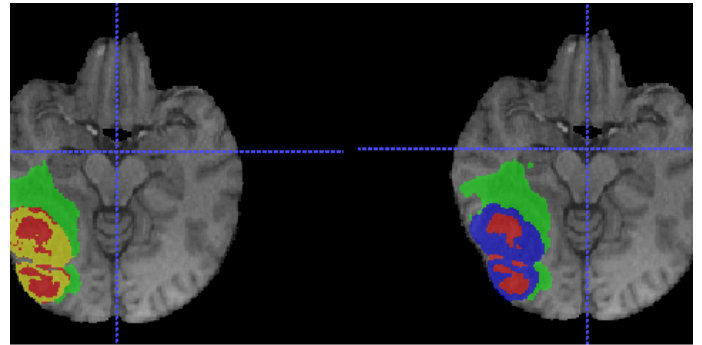
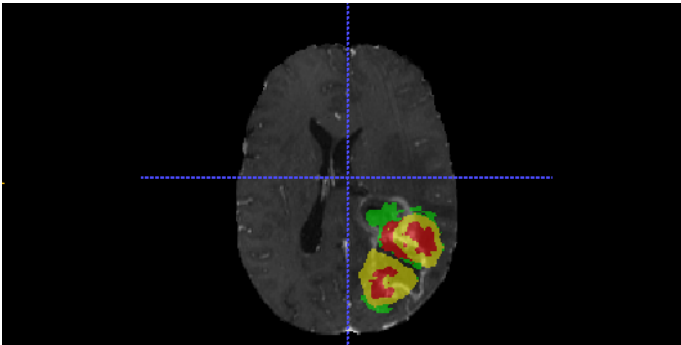
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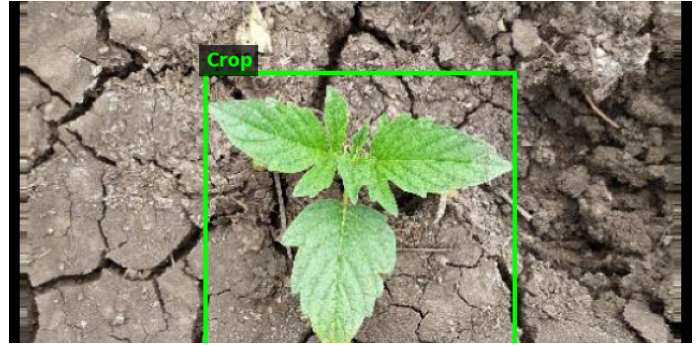
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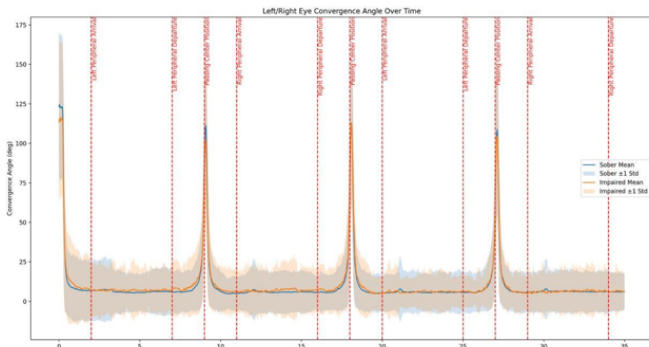
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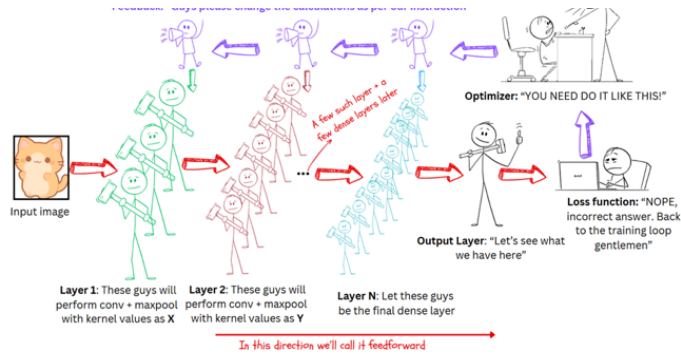
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